

Project Proposal: Close Field Agricultural Vision Foundation Models Trained with Self-Supervised Learning

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Abstract

This project will develop an agricultural vision foundation model that transfers across crops, field conditions, and sensing modalities. The work will continue self-supervised pre-training of official ImageNet-initialized Vision Transformers (ViTs) on a curated agricultural corpus and compare masked image modeling (MIM) and DINO-style self-distillation. A learnable 1×1 spectral adapter will allow the same RGB-pretrained ViT to accept RGB images, GeoTIFF tiles, and multispectral arrays. Success will be assessed through evaluation on downstream tasks.

1. Background and Significance

Agricultural computer-vision systems are commonly trained for a single task and dataset, for example leaf-disease classification or weed recognition. This setting does not reflect deployment: image appearance changes with crop species, phenology, illumination, soil background, farm, season, geography, and sensor. Labels for these conditions are costly to obtain, while useful data are fragmented across small, heterogeneous public datasets. A foundation model that learns transferable agricultural representations could reduce annotation requirements and support downstream classification, detection, segmentation, retrieval, and monitoring applications.

The proposed work addresses two technical limitations. First, generic ViT checkpoints are trained primarily on natural RGB imagery and may not represent agricultural symptoms, morphology, or field context effectively. Second, agricultural sensing includes close-range RGB photographs as well as georeferenced multispectral observations, whereas standard pretrained ViTs assume three RGB input channels. The project therefore combines agricultural continual pre-training with a lightweight interface for sensor adaptation.

2. Project Objectives and Hypotheses

Objective 1: Create a agricultural pretraining corpus with documented composition, licenses.

Objective 2: Pretrain official ViT-S and ViT-B backbones

using MIM, DINO-style self-distillation.

Objective 3: Evaluate whether a small spectral adapter enables effective RGB-to-multispectral transfer without replacing the pretrained ViT.

Objective 4: Establish transfer performance under realistic source, crop, geography, season, and sensor shift.

The project tests three hypotheses:

H1: agricultural continual pretraining will improve low-label and held-out-domain transfer relative to an unadapted ImageNet ViT.

H2: reconstruction and self-distillation will provide complementary benefits, making a sequential MIM-to-DINO schedule a competitive candidate.

H3: a trainable 1×1 spectral adapter will transfer RGB-pretrained representations to genuine multispectral inputs more effectively than an RGB-only baseline, while adding negligible parameters.

3. Data, Scope, and Governance

The initial corpus measured **412,454** images in **13** downloaded datasets: 400,359 JPG, 11,079 PNG, and 1,010 JPEG files.

Before training, each source will be recorded with its license, citation, version, retrieval date, source URL, modality, channel metadata, spatial metadata, and exclusions. Exact and near-duplicate images will be removed or grouped before any split is defined. The release will provide code, manifests, and permitted metadata rather than redistribute sources whose licenses prohibit redistribution. Claims about multimodal or sensor-general learning will be made only after real GeoTIFF or multispectral data are incorporated and evaluated.

4. Research Design and Methodology

Model initialization and sensor adaptation. The project will use official `timm` ViT-S and ViT-B checkpoints initialized from ImageNet, MAE, DINOv2, or DINOv3 weights, rather than randomly initialized transformers. A learnable 1×1 convolution will project an arbitrary C-band input into the three-channel interface expected by the pretrained ViT. For RGB input, the adapter is identity initialized. For

multispectral input, the available bands are mapped to the RGB interface and additional band weights begin at zero while remaining trainable. This preserves the pretrained backbone while creating a clear ablation against direct patch-embedding expansion.

Continual-pretraining objectives. MIM will train the encoder to reconstruct masked patches, encouraging local structural and textural features. DINO-style training will use a student–teacher self-distillation objective with an exponential-moving-average teacher and multi-crop augmentations to encourage view-invariant representations. The core comparison will include ImageNet-only transfer, MIM-only and DINO-only backbone.

5. Evaluation Plan and Success Criteria

Experiments will begin with ViT-S on a single RTX 4090 (24 GB), using BF16 mixed precision, gradient accumulation, and gradient checkpointing. Stable recipes will be scaled to ViT-B and, where feasible, ViT-L. After curation, sampling will be source balanced so that large species collections do not dominate disease, weed, or multispectral examples.

Representation quality will be evaluated through linear probing, few-shot adaptation, and full fine-tuning. Classification will report accuracy and macro-F1; compatible detection and segmentation datasets will additionally use mAP and IoU. The primary evaluation protocol will hold out entire sources, crops, farms, geographies, seasons, or sensors. Random image-level splits will serve only as secondary diagnostics. Required baselines are an unadapted ImageNet ViT, RGB-only pretraining, MIM-only and DINO-only training, frozen and trainable adapters, and direct patch-embedding expansion where feasible.

The project will be successful if it demonstrates consistent improvement over generic ViT initialization across multiple transfer settings, stronger performance under limited labels, and credible gains from real multispectral data once available. Supporting evidence will include per-domain metrics, label-efficiency curves, repeated-seed mean and standard deviation, training curves, MIM reconstructions, and DINO student–teacher similarity visualizations.

6. Deliverables, Risks, and Work Plan

The project will produce: (1) a versioned data catalog with provenance and duplicate reports; (2) reproducible MIM and DINO-style pretraining configurations for official ViTs; (3) spectral-adaptor ablations for RGB and multispectral inputs; (4) domain-shift transfer benchmarks.

The principal risks are dataset leakage, RGB-heavy sampling bias, and inadequate multispectral coverage. These risks will be managed through duplicate audits, source-disjoint evaluation, modality-aware manifests, and an explicitly scoped conclusion if multispectral acquisition remains

incomplete.

Weeks 1–2: freeze corpus version 1, complete provenance and duplicate audits, establish ImageNet ViT baselines, and select stable ViT-S recipes.

Weeks 3–4: acquire and tile GeoTIFF/multispectral sources; conduct sensor-adaptor and objective ablations; evaluate domain-held-out transfer.

Weeks 5–6: scale the selected recipe, repeat critical experiments, prepare results, and release the code, manifests, configurations, and permitted checkpoints for submission to a computer-vision or agricultural-informatics venue.

7. References

[1] A. Dosovitskiy et al. *An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale*. ICLR, 2021. [2] K. He et al. *Masked Autoencoders Are Scalable Vision Learners*. CVPR, 2022. [3] M. Caron et al. *Emerging Properties in Self-Supervised Vision Transformers*. ICCV, 2021. [4] M. Oquab et al. *DINOv2: Learning Robust Visual Features without Supervision*. TMLR, 2024.